

Gramática en Notación Backus-Naur (BNF)

```
<inicio> ::= <funcion_main>

<funcion_main> ::= "func" "main" "(" ")" "{" <instrucciones> "}"

<instrucciones> ::= <instrucciones> <instruccion>
                  | ε

<instruccion> ::= <declaracion>
                  | <impresion>
                  | <if_instr>
                  | <for_instr>
                  | <incremento>
                  | <decremento>
                  | "{" <instrucciones> "}"
                  | "break"
                  | "continue"

<incremento> ::= ID "++"

<decremento> ::= ID "--"

<if_instr> ::= "if" <expresion> "{" <instrucciones> "}"
            | "if" <expresion> "{" <instrucciones> "}" "else" "{" <instrucciones> "}"

<for_instr> ::= "for" <expresion> "{" <instrucciones> "}"
            | "for" <declaracion> ";" <expresion> ";" <actualizacion_for> "{" <instrucciones> "}"

<declaracion> ::= "var" ID <tipo> "=" <expresion>
                | "var" ID <tipo>
                | ID ":" <expresion>
                | ID "=" <expresion>
                | ID "+" <expresion>
                | ID "-" <expresion>

<actualizacion_for> ::= ID "+" <expresion>
                    | ID "-" <expresion>
                    | ID "=" <expresion>
                    | ID "++"
                    | ID "--"

<tipo> ::= "int"
        | "float64"
        | "string"
        | "bool"
        | "rune"

<impresion> ::= "fmt.Println" "(" <lista_print> ")"

<lista_print> ::= <lista_print> "," <expresion>
               | <expresion>

<expresion> ::= <expresion> "+" <expresion>
               | <expresion> "-" <expresion>
               | <expresion> "*" <expresion>
               | <expresion> "/" <expresion>
               | <expresion> "%" <expresion>
               | <expresion> ">" <expresion>
               | <expresion> "<" <expresion>
               | <expresion> ">=" <expresion>
               | <expresion> "<=" <expresion>
               | <expresion> "==" <expresion>
               | <expresion> "!=" <expresion>
               | <expresion> "&&" <expresion>
               | <expresion> "||" <expresion>
               | "!" <expresion>
               | "-" <expresion>
               | "(" <expresion> ")"
               | "strconv.Atoi" "(" <expresion> ")"
               | "strconv.ParseFloat" "(" <expresion> "," <expresion> ")"
               | "reflect.TypeOf" "(" <expresion> ")" ".String" "(" ")"
               | ENTERO
               | DECIMAL
               | CADENA
```

```
| CHARACTER
| "true"
| "false"
| "nil"
| ID
```